

```
import cv2
import numpy as np
from google.colab import files
from google.colab.patches import cv2_imshow

# -----
# Step 1: Upload an image from your computer
# -----
print("Please upload an image file:")
uploaded = files.upload()

# Get the uploaded image path automatically
image_path = list(uploaded.keys())[0]

# -----
# Step 2: Read the image using OpenCV
# -----
image = cv2.imread(image_path)

if image is None:
    print("Error: Could not read image!")
else:
    # -----
    # Step 3: Define a structuring element (kernel) and Dilate
    # -----
    # Create a 5x5 matrix of ones as the structuring element
    kernel = np.ones((5, 5), np.uint8)

    # Perform Dilation
    dilated_image = cv2.dilate(image, kernel, iterations=1)

    # -----
    # Step 4: Display the results in Colab
    # -----
    print("\n--- Original Image ---")
    cv2_imshow(image)

    print("\n--- Dilated Image ---")
    cv2_imshow(dilated_image)
```

Please upload an image file:
 exp 4 image.jpg
exp 4 image.jpg(image/jpeg) - 46154 bytes, last modified: 20/7/2026 - 100% done
Saving exp 4 image.jpg to exp 4 image.jpg

--- Original Image ---



--- Dilated Image ---

